Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

SVD-ROM: Scalable Reduced Order Modeling of Weather and Climate Data Using the Singular Value Decomposition

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Received 31 Mar 2026

Keywords: climate informatics, reduced order modeling, singular value decomposition, dynamic mode decomposition

1. Introduction and objectives

Despite their high dimensionality, datasets in fluid dynamics, weather, and climate often exhibit low-rank structure, where a small number of dominant patterns explain most of the variability. Singular Value Decomposition (SVD)-based methods provide efficient and interpretable tools for dimensionality reduction in such systems. Dynamic Mode Decomposition (DMD) extends this framework to time-resolved data by extracting coherent spatio-temporal structures and their associated dynamics [Schmid(2022)]. This enables the construction of low-dimensional, interpretable emulators of complex dynamical systems. However, its adoption is limited by the practical challenges of working with modern geophysical data volumes. In this work, we present SVD-ROM [Salvador-Jasin(2026)], an open-source Python package for SVD-based reduced order modeling, including DMD, designed to operate efficiently on datasets such as ERA5 using parallel and out-of-core computation on standard hardware.

2. Methods and data

SVD-ROM enables the application of DMD to datasets larger than memory through two key components: randomized SVD and optimized DMD. The randomized SVD [Halko(2011)] provides an approximate matrix factorization that is both computationally efficient and robust to noise compared to deterministic approaches. When combined with a communication-efficient, parallel QR factorization for tall-and-skinny matrices (TSQR) [Benson(2013)], it enables scalable low-rank approximations of arbitrarily large, moderately wide matrices [Tepper(2016)]. Once the SVD factors have been computed, an optimized and noise-aware formulation of DMD [Askham(2018)] is applied to the reduced representation, allowing the dominant spatio-temporal modes to be extracted efficiently without revisiting the full dataset. In practice, this enables the computation of DMD on ERA5 subsets of approximately 60 GB on a 32 GB, 10-core laptop in under 20 minutes, comparable to the cost of computing a sliding-window climatology on the same dataset. SVD-ROM employs Dask for parallel, out-of-core computation, and the Zarr and netCDF formats for compressed chunked storage and fast data access.

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3. Preliminary results

DMD models were fitted to two 60 GB subsets of the 0.25° ERA5 dataset, each consisting of a 10-year time series of global atmospheric temperature at 850 hPa with 6-hourly resolution. The first subset spans January 2010 to December 2019, while the second spans July 2010 to June 2020. Figure 1 shows the Root Mean Square Error (RMSE) and Anomaly Correlation Coefficient (ACC) for reconstructions of the final two months of training data and for 45-day forecasts from both models. The results are compared against short-term climatology (2010-2019) and long-term climatology (1990-2019). DMD forecasts match the climatology RMSE throughout the forecast horizon, indicating that the model captures variability at a level comparable to baseline statistical benchmarks. While the ACC is low, it remains positive, indicating that the forecasts retain limited but non-random skill over extended lead times. Considering the fitted model occupies under 1.5 GB on disk, the results highlight the capability of DMD to provide a highly compressed, low-dimensional operator that captures the dominant spatio-temporal dynamics while enabling efficient storage, analysis, and forecasting.

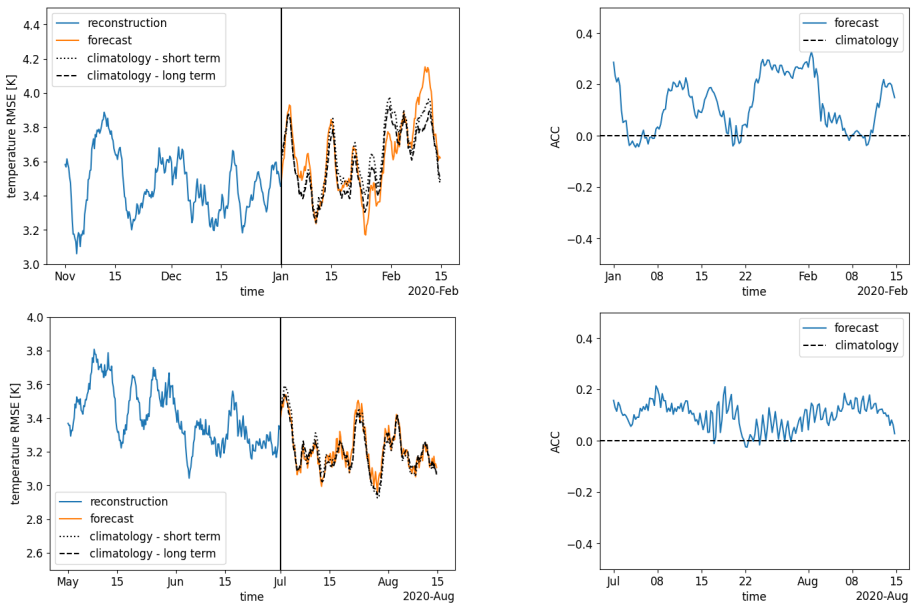

Figure 1. *RMSE of atmospheric temperature at 850 hPa for a two month reconstruction and a 45-day forecast (left), and ACC of the 45-day forecast (right), for winter (top) and summer (bottom) 2020.*

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